

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCPE705E - APPLIED HYDROLOGY AND FLOOD CONTROL

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 **The hyetograph represents:**
 - a) Cumulative rainfall over time
 - b) Rainfall intensity versus time
 - c) Area–depth relationship
 - d) Frequency of peak discharge
- 2 **A mass rainfall curve is obtained by plotting:**
 - a) Rainfall intensity vs. duration
 - b) Accumulated rainfall vs. time
 - c) Rainfall depth vs. area
 - d) Rainfall frequency vs. magnitude
- 3 **IDF curves are used to determine:**
 - a) Rainfall temperature variation
 - b) Maximum rainfall intensity for a given duration
 - c) Evaporation losses
 - d) Depth–area relation of storms
- 4 **The major factor influencing runoff generation is:**
 - a) Soil infiltration capacity
 - b) Longitude of region
 - c) Ocean currents
 - d) Atmospheric pressure
- 5 **Base flow in a hydrograph represents:**
 - a) Rainfall excess
 - b) Groundwater contribution to streamflow
 - c) Direct runoff
- 6 **The unit hydrograph is defined for:**
 - a) 1 cm of effective rainfall of specified duration
 - b) 10 cm rainfall for any duration
 - c) Any rainfall, any duration
 - d) Only flood hydrograph
- 7 **The rational formula is commonly used for estimating peak flow in:**
 - a) Large river basins
 - b) Medium-sized watersheds
 - c) Small urban catchments
 - d) Groundwater aquifers
- 8 **Gumbel's method is associated with:**
 - a) Sediment analysis
 - b) Frequency analysis of extreme events
 - c) Base flow separation
 - d) Estimation of unit hydrograph

- 9 **SPF and MPF methods are used to estimate:**
 a) Groundwater recharge
 b) Water quality index
 c) Design flood
 d) Evaporation losses
- 10 **Muskingum routing is a type of:**
 a) Distributed rainfall model
 b) Hydraulic routing method
 c) Hydrologic channel routing
 d) Lumped model only
- 11 **Hydrological reservoir routing mainly determines:**
 a) Temperature changes in water
 b) Storage–outflow relationship
 c) Soil infiltration
 d) Land use classification
- 12 **Floodplain zoning refers to:**
 a) Allocating urban zones for industries
 b) Restricting or regulating land use in flood-prone areas
 c) Constructing floodwalls
 d) Mapping agricultural zones

Q2. Solve the following.

- A) Explain the construction and applications of a hyetograph. How is it useful in hydrological analysis? 6
- B) Describe the concept of rainfall frequency analysis. Explain the steps involved in determining the return period of a point rainfall. 6

Q3. Solve the following.

- A) Discuss the various factors that influence runoff generation in a watershed. Illustrate with suitable examples. 6
- B) Explain the procedure for base flow separation from a flood hydrograph. What is its significance in hydrological studies? 6

Q4. Solve Any Two of the following.

- A) Describe different types of floods and their causative factors. How do climatic and physiographic conditions influence flood occurrence? 6
- B) Explain the rational method of peak flow estimation. Discuss the assumptions, limitations, and applicability of the method. 6
- C) What is flood frequency analysis? Describe the steps involved in Gumbel's method for estimating design floods. 6

Q5. Solve Any Two of the following.

- A) Discuss SPF and MPF estimation approaches. How are these empirical methods used in determining design flood? 6
- B) Explain the Muskingum method of flood routing. Derive the basic routing equation and discuss the role of storage parameters. 6
- C) Compare lumped and distributed hydrological models for flood estimation. What are their advantages, limitations, and applications? 6

Q6. Solve Any Two of the following.

- A) Describe the structural and non-structural methods of flood control. Give examples of each and explain their importance in modern flood management. 6
- B) Explain the procedure of hydrologic reservoir routing. How are storage–indication curves used for outflow calculations? 6
- C) Discuss the role of remote sensing and GIS in flood disaster monitoring, forecasting, and mitigation. Provide practical examples. 6

*** End ***